XGBoost Results — Ganymede Gas Production Forecasting

Supplementary Tables for M004

March 2026

Table 1: Ganymede forecasting (held-out test, multi-well, MMSCF). Best values per column in bold. FM = foundation model (zero-shot). Tree = gradient boosted trees (no neural network).

Model	**Metric**	***h*=7**	***h*=14**	***h*=30**	***h*=90**
TimesFM$^{\mathrm{FM}}$	R^2_{prod}	**.826**	.763	.656	.413
	MAE	.740	.868	.949	1.02
TiRex$^{\mathrm{FM}}$	R^2_{prod}	.822	**.775**	**.699**	**.565**
	MAE	**.658**	**.749**	**.804**	**.831**
Chronos$^{\mathrm{FM}}$	R^2_{prod}	.734	.668	.584	.361
	MAE	.739	.835	.886	.995
PatchTST	R^2_{prod}	.556	.466	.211	−.382
	MAE	1.77	1.98	2.14	2.48
LSTM	R^2_{prod}	.195	.534	.303	.020
	MAE	1.77	1.69	1.84	2.08
XGBoost$^{\mathrm{Tree}}$	R^2_{prod}	−.031	−.465	−.415	−1.25
	MAE	1.93	2.45	2.47	2.77
DeepONet	R^2_{prod}	−40.1	−4.35	−5.01	−4.32
	MAE	9.53	4.12	4.51	3.84

Table 2: XGBoost held-out test metrics across all horizons (multi-well, MMSCF).

Metric	***h*=7**	***h*=14**	***h*=30**	***h*=90**
MAE	1.928	2.447	2.468	2.768
RMSE	3.196	3.792	3.650	4.058
R^2	−0.078	−0.574	−0.664	−1.720
R^2_{prod}	−0.031	−0.465	−0.415	−1.252
MASE	4.381	2.853	2.905	3.836

Table 3: XGBoost 3-fold expanding window CV aggregates (multi-well, mean ± std).

Metric	***h*=7**	***h*=14**	***h*=30**	***h*=90**
MAE	3.10 ± 1.20	3.26 ± 1.23	3.81 ± 0.73	3.80 ± 1.50
RMSE	6.11 ± 2.88	6.33 ± 2.62	7.39 ± 1.95	7.43 ± 3.22
R^2_{prod}	0.69 ± 0.24	0.60 ± 0.36	-0.23 ± 1.52	-0.15 ± 1.34
MASE	1.88 ± 0.94	1.76 ± 0.87	2.10 ± 0.61	2.30 ± 0.85

Table 4: XGBoost fold-level CV metrics (multi-well). 3-fold expanding window.

h	Fold	MAE	RMSE	R^2	R^2_{prod}	MASE
	0	1.426	2.037	0.366	0.354	3.208
7	1	3.686	8.294	0.795	0.840	1.224
	2	4.195	7.988	0.851	0.891	1.201
	0	1.622	2.380	0.314	0.237	1.971
14	1	3.718	8.136	0.712	0.790	1.289
	2	4.434	8.468	0.818	0.773	2.013
	0	3.019	5.145	0.191	0.129	1.514
30	1	3.873	7.819	0.652	0.403	2.067
	2	4.527	9.197	0.643	−1.217	2.734
	0	2.102	3.561	−2.038	−2.038	1.228
90	1	5.747	10.99	0.800	0.800	3.369
	2	3.561	7.741	0.793	0.793	2.294

Table 5: XGBoost per-well held-out test metrics, h=7.

Well	MAE	RMSE	R^2	R^2_{prod}	MASE
49/22-Z01Z	0.372	0.561	−80.33	−31.34	9.229
49/22-Z02Z	2.122	2.744	0.048	−0.163	2.903
49/22-Z04	0.767	1.002	0.076	−0.244	2.291
49/22-Z05Z	0.713	0.917	−3.580	−4.709	5.368
49/22-Z06	3.099	4.320	−20.73	−5.170	12.530
49/22-Z07	1.734	2.343	0.240	0.292	1.979
49/22-Z08	1.915	2.591	−4.011	−1.101	2.185

Table 6: XGBoost per-well held-out test metrics, h=14.

Well	MAE	RMSE	R^2	R^2_{prod}	MASE
49/22-Z01Z	0.301	0.427	−67.61	−28.00	7.701
49/22-Z02Z	2.562	3.411	−0.373	−0.453	1.289
49/22-Z04	0.863	1.080	−0.018	−0.383	1.434
49/22-Z05Z	0.719	0.906	−2.741	−4.528	2.526
49/22-Z06	4.310	5.891	−220.7	−46.53	19.595
49/22-Z07	1.974	2.639	0.003	0.134	1.233
49/22-Z08	2.270	2.668	−6.350	−1.008	3.412

Table 7: XGBoost per-well held-out test metrics, h=30.

Well	MAE	RMSE	R^2	R^2_{prod}	MASE
49/22-Z01Z	0.353	0.540	−124.9	−48.34	10.472
49/22-Z02Z	3.232	4.227	−1.015	−1.077	1.575
49/22-Z04	1.134	1.380	−0.754	−0.481	1.845
49/22-Z05Z	0.692	0.883	−2.021	−4.154	2.453
49/22-Z06	4.297	5.475	−412.0	−71.77	35.261
49/22-Z07	2.399	3.124	−0.438	0.044	1.571
49/22-Z08	3.340	3.491	−102.5	−6.832	25.989

Table 8: XGBoost per-well held-out test metrics, h=90.

Well	**MAE**	**RMSE**	$\boldsymbol{R^2}$	$\boldsymbol{R^2_{\text{prod}}}$	**MASE**
49/22-Z01Z	0.390	0.590	−145.4	−50.10	12.073
49/22-Z02Z	6.537	8.903	−8.059	−6.449	3.824
49/22-Z04	1.548	1.847	−2.628	−0.814	3.153
49/22-Z05Z	0.681	0.910	−1.900	−1.928	3.398
49/22-Z06	3.573	4.523	−6549.7	−511.9	307.8
49/22-Z07	2.775	3.443	−0.829	−0.185	2.287
49/22-Z08	0.179	0.363	−1.170	−1.083	2.822

Table 9: XGBoost R^2_{prod} per well across all horizons (held-out test).

Well	$\boldsymbol{h}$**=7**	$\boldsymbol{h}$**=14**	$\boldsymbol{h}$**=30**	$\boldsymbol{h}$**=90**
49/22-Z01Z	−31.34	−28.00	−48.34	−50.10
49/22-Z02Z	−0.163	−0.453	−1.077	−6.449
49/22-Z04	−0.244	−0.383	−0.481	−0.814
49/22-Z05Z	−4.709	−4.528	−4.154	−1.928
49/22-Z06	−5.170	−46.53	−71.77	−511.9
49/22-Z07	0.292	0.134	0.044	−0.185
49/22-Z08	−1.101	−1.008	−6.832	−1.083

Table 10: XGBoost MAE per well across all horizons (held-out test, MMSCF).

Well	$\boldsymbol{h}$**=7**	$\boldsymbol{h}$**=14**	$\boldsymbol{h}$**=30**	$\boldsymbol{h}$**=90**
49/22-Z01Z	0.372	0.301	0.353	0.390
49/22-Z02Z	2.122	2.562	3.232	6.537
49/22-Z04	0.767	0.863	1.134	1.548
49/22-Z05Z	0.713	0.719	0.692	0.681
49/22-Z06	3.099	4.310	4.297	3.573
49/22-Z07	1.734	1.974	2.399	2.775
49/22-Z08	1.915	2.270	3.340	0.179